

PROFILE

Platform and site reliability engineer with about eight years of experience building and operating cloud-native application platforms across AWS EKS, Azure, Kubernetes, and OpenShift.

Combines platform software, release automation, observability, access control, and incident response to move services into reliable production operation.

SELECTED EVIDENCE

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|--------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| AWS EKS agent platform with automated domains and authorized-only access | Production delivery and reliability for a 30,000-employee AI service | Kubernetes delivery, observability, and incident response across cloud environments |
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CORE SKILLS

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|---------------------|------------------------------------------------------------------------------------------|
| Platform            | AWS EKS, Kubernetes, OpenShift, Azure, Docker, Helm/Helmfile, Kubernetes Operators       |
| Delivery Automation | ArgoCD, GitOps, GitHub Actions, Jenkins, CI/CD, Bash-based cluster provisioning          |
| Reliability         | Grafana, Prometheus, OpenTelemetry, high availability, auto-scaling, incident response   |
| Software & Security | Python, Bash, Linux, API integration, authentication, authorization, security governance |

EXPERIENCE

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| Samsung Electronics, AX Development Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Seoul, South Korea    |
| SRE / AI Platform Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jan. 2026 - Present   |
| <ul style="list-style-type: none"><li>- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards.</li><li>- Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance.</li><li>- Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration.</li><li>- Operated an Azure-based generative AI service and drove its global rollout, covering production readiness, incident response, and post-launch stabilization.</li></ul> |                       |
| Samsung Research                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Seoul, South Korea    |
| SRE / AI Platform Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jul. 2024 - Dec. 2025 |
| <ul style="list-style-type: none"><li>- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters.</li><li>- Owned releases across development, staging, and production while integrating application delivery with security and operational requirements.</li><li>- Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile.</li><li>- Deployed Grafana, Prometheus, and OpenTelemetry observability for shared service visibility and operational diagnosis.</li></ul>                                                                                                                         |                       |
| Samsung Electronics, Network Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Seoul, South Korea    |
| SRE & Software Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Jan. 2021 - Jun. 2024 |
| <ul style="list-style-type: none"><li>- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM.</li><li>- Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling.</li><li>- Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers.</li><li>- Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.</li></ul>                                                                                                                 |                       |

EXPERIENCE (CONTINUED)

TmaxOS & TmaxCloud

Software & Platform Engineer

Gyeonggi, South Korea  
Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Kubernetes Cluster Automation

Designed and developed a Bash-based one-click provisioning system for repeatable developer Kubernetes environments.

Centralized Observability Stack

Deployed Grafana, Prometheus, and OpenTelemetry for shared service visibility, dashboards, and operational diagnosis.

High-throughput Packet Mirroring

Developed a resource-aware Linux application measured at up to 130,000 packets per second.

Internal AX Agent Platform

Built an AWS EKS application platform with automatic domains, authorized-only access, enterprise security controls, and compatibility with an existing LLM Gateway API.

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering

Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

Mar. 2017 - Feb. 2019

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics

Undergraduate research: distributed systems for AI model training with TensorFlow

Mar. 2012 - Feb. 2017

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPic IH (Intermediate High), Dec. 2025.